

NEXUS Engineering Physics: Module 1

Vector Calculus and Electrostatic Fields

1.1 Vector Fields and Flux

A vector field assigns a vector to every point in space. The flux of an electric field E through a differential surface element dA is defined as $d\Phi = E \cdot dA$. For a closed surface S bounding a volume V , the total outward flux is given by the surface integral $\Phi = \oint_S E \cdot dA$.

1.2 Gauss's Divergence Theorem

Gauss's Divergence Theorem is a fundamental theorem of vector calculus that relates the flux of a vector field across a closed boundary surface to the volume integral of its divergence.

Statement: The outward flux of a continuously differentiable vector field F through a closed surface S bounding a volume V equals the volume integral of the divergence of F over V .

Mathematically: $\oint_S F \cdot dA = \int_V (\text{div } F) dV$

In physical terms, divergence measures the rate at which field lines expand from a point source. If the divergence is positive at a point, that point behaves like a source. If the divergence is negative, the point behaves like a sink.

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1.3 Physical Application to Gauss's Law in Electrostatics

In electrostatics, Gauss's Law states that the net electric flux through any closed Gaussian surface is proportional to the total electric charge enclosed by that surface:

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$$

Applying the Divergence Theorem, the left-hand side transforms to the volume integral of $\text{div } \mathbf{E}$. Equating integrands for an arbitrary volume V yields the differential form of Gauss's Law:

$$\text{div } \mathbf{E} = \rho / \epsilon_0$$

where ρ is the volume charge density and ϵ_0 is the permittivity of free space. This differential form is one of Maxwell's equations and is the local statement of Gauss's law.